Bihar Engineering University, Patna

B.Tech 7th Semester Examination, 2024

Course: B.Tech
Code: 103715

Subject: Electrical Materials

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) X-ray diffraction is used to determine:
(i) Electrical conductivity
(ii) Crystalline structure
(iii) Magnetic properties
(iv) Optical reflectivity

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(b) Which of the following is NOT an imperfection in solids?
(i) Dislocations
(ii) Grain Boundaries
(iii) Atomic Bonds
(iv) Vacancies

(c) The material commonly used in piezoelectric applications is:
(i) Ferrite
(ii) Quartz
(iii) Aluminum
(iv) PVC

(d) Which of the following properties is NOT a characteristic of insulating materials?
(i) High volume resistivity
(ii) High electrical conductivity
(iii) High electrical conductivity
(iv) Low dielectric loss

(e) Which insulating material is commonly used in transformers?
(i) Bitumen
(ii) Mineral Oil
(iii) PVC
(iv) Epoxy Resins

(f) A major advantage of thermoplastics over thermosetting plastics is:
(i) Higher temperature resistance
(ii) Ability to be remoulded
(iii) Greater brittleness
(iv) Lower strength

(g) What is the primary carrier in an intrinsic semiconductor?
(i) Holes
(ii) Both electrons and holes
(iii) Electrons
(iv) Phonons

(h) Which optical material is widely used in fiber-optic communications?
(i) Quartz
(ii) Copper
(iii) Silicon
(iv) Gallium arsenide

(i) Superconductivity is characterized by:
(i) Infinite resistance
(ii) High temperature stability
(iii) Zero resistance
(iv) High dielectric constant

(j) Which property is defined as the ability of insulating material to resist electric breakdown?
(i) Dielectric Strength
(ii) Volume Resistivity
(iii) Thermal Conductivity
(iv) Surface Resistance

Q.2 (a) Explain the importance of different types of inter atomic bonding in electrical materials. [7]

(b) Discuss the classification of materials based on their electrical and structural properties. [7]

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Q.3	(a)	What is X-ray diffraction? Explain its role in determining the structure of solids?	[7]
	(b)	Discuss imperfections and defects in crystalline solids.	[7]
Q.4	(a)	Explain the general properties of insulating materials and their role in electrical applications.	[7]
	(b)	Describe the different types of plastics used in electrical insulation.	[7]
Q.5	(a)	Explain the concept of electronic and ionic conduction in materials.	[7]
	(b)	Discuss the band theory and its application in semiconductors.	[7]
Q.6	(a)	Explain the optical properties of metals and non-metals. Describe the basic principles of luminescence and its applications.	[7]
	(b)	Discuss the different types of magnetism observed in materials.	[7]
Q.7	(a)	What are superconductors? Discuss their fundamental properties.	[7]
	(b)	Explain the Meissner effect and its role in superconductivity.	[7]
Q.8	(a)	Describe the domain theory of ferromagnetism. Explain the significance of hysteresis loss in transformer cores.	[7]
	(b)	Describe the applications of superconductors in modern technology. Discuss the impact of material recycling on sustainability.	[7]
Q.9	(a)	What are biodegradable and bio renewable polymers? Give examples.	[7]
	(b)	How do nanowires differ from conventional materials? Discuss their applications.	[7]

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